Name→Diya Goel Roll No.→ 2401010179

TOPIC DATE

# Database Management System [ETCS 204]
## Assignment - 2

Section - A [Fill in the blanks]

Ques-1 Fill in the blanks:

1- The Entity Relationship (ER) Diagram visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

2- In the Relational model, data is organized into tables, with relationships between entities defined through foreign key constraints.

3- The Object-oriented model treats data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions.

4- Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Section - B [Case Study Based]

Ques-2 In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professionals and patient records.

Data abstraction means hiding complex database details and showing only important information to users. In a healthcare system, it helps doctors, nurses, and staff access patient records easily.

TOPIC _____ DATE _____

There are three levels of data abstraction:-

Physical level - Describes how data is stored in the database. Users do not interact with this level.

(2) Logical level - Defines what data is stored, such as patient details, doctor information and appointments and the relationships between them.

(3) View level - Shows only required information to different users. For example, doctors can see medical history, nurses can see patient vitals, and receptionists can see appointment details.

Thus, data abstraction makes the healthcare system simple, secure and efficient.

Ques-3 Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Benefits-:

(1) Improves data security and privacy
(2) Makes the system easy to use for healthcare staff.
(3) Provides data independence, so changes in databases structure do not affect users.
(4) Helps in better data management.

TOPIC _____ DATE _____

Challenges:

1. Designing abstraction layers can be complex
2. It may cause slight performance overhead
3. Implementation can be costly
4. Staff may require training to use the system.

Section-C [Simulation - based]

Q-3 Design a database for a university using the ER model.

In a university, the main entities are student, Course and Professor. These entities store important information about students, courses and teachers.

Entities and Attributes →

1. Student
   - Student_ID [Primary key]
   - Name
   - Gender
   - Age
   - Email
2. Course
   - Course_ID [Primary key]
   - Course_Name
   - Credits
   - Department
3. Professor
   - Professor_ID [Primary key]
   - Professor_Name
   - Department
   - Specialization

TOPIC _____ DATE _____

Relationships →

① Enrollement Relationship →

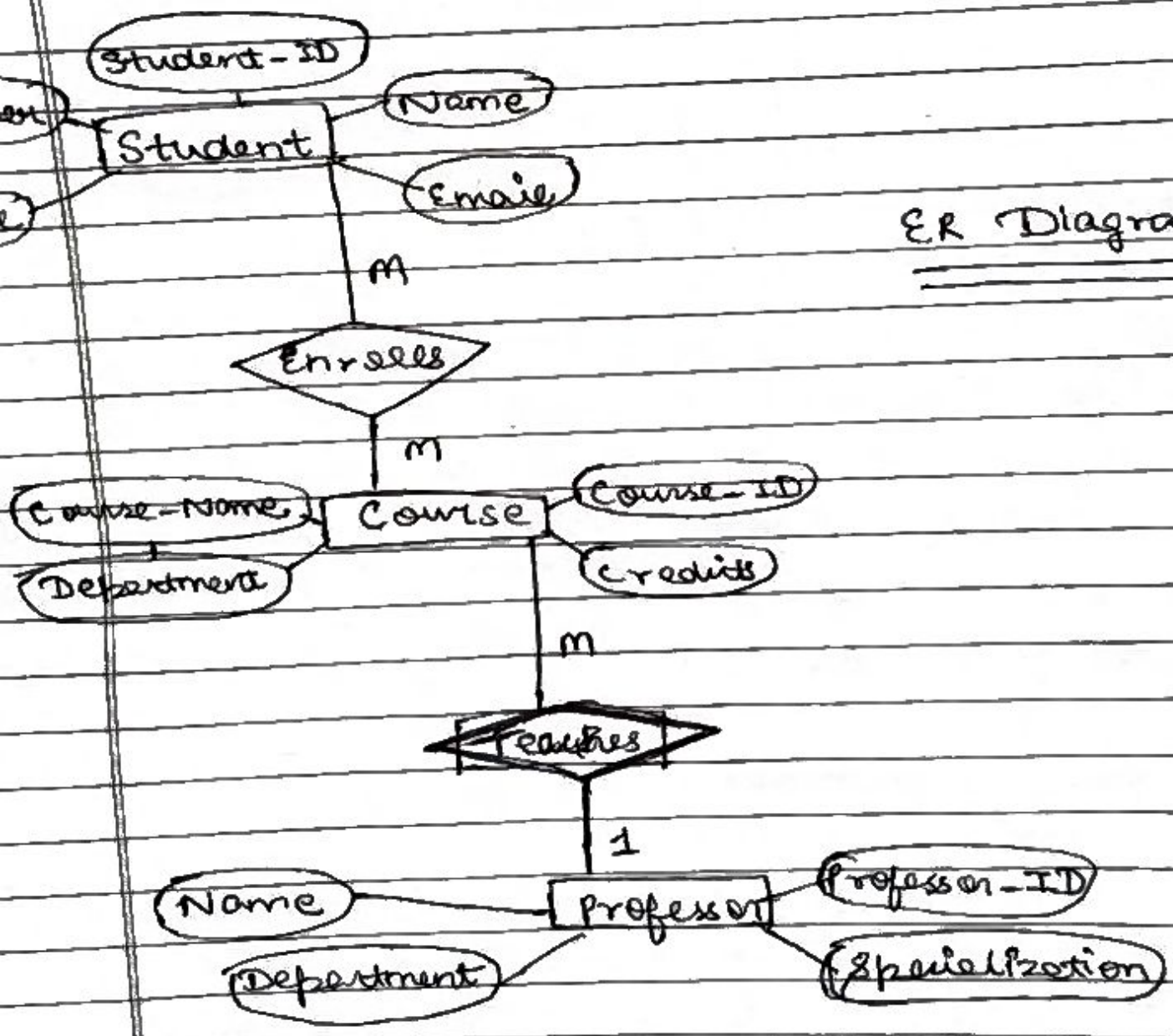
A student can enroll in many courses, and a course can have many students.

This is a many-to-many (M:N) relationship.

② Teacher Relationship →

A professor teaches courses and a course is taught by professor.

This is a one-to-many (1:M) relationship.




ER Diagram